

SPIRE Structured Note Pricer

model: "spire"

Capital-protected notes — collateral + swap decomposition

Description

Prices capital-protected and partial-protection structured notes constructed as a collateral bond plus a swap overlay. The collateral bond (typically a zero-coupon or low-coupon government or SSA bond) provides the capital guarantee; the swap adjustments price the structured coupon and performance features. Supports fixed, autocallable, and Bermudan-callable structures.

Pricing Methodology

Note PV = Collateral PV + Swap adjustments – Issuer spread PV

The collateral bond is priced using the Hull-White DCF engine against its own discount curve. Swap adjustments represent the difference between the structured coupon (paid to the investor) and the natural yield of the collateral.

For autocallable structures, Monte Carlo simulation (GBM) determines the probability of early redemption at each observation date.

Key Formulas

Note PV = Collateral PV + Swap adjustments – Issuer spread PV

Collateral PV = $\sum \text{collateral_CF}(t) \times \text{DF_collateral}(t)$

For autocall: $\text{PV} = E^Q[\text{CF on earliest autocall date} \mid \text{GBM paths}]$

Required Fields

Field	Type	Description
instrument_id	string	ISIN or internal identifier
evaluation_date	string	Pricing date
coupon_structure	string	Must be "fixed"
note_notional	float	Notional of the note
coupon_frequency	string	Annual Semiannual Quarterly Monthly
calendar	string	TARGET UnitedStates
collateral	object	Collateral bond definition (see below)
collateral.principal_amount	float	Collateral principal
collateral.coupon_rate	float	Collateral coupon
collateral.maturity_date	string	Collateral maturity

Optional Fields

Field	Type	Description
credit_spread_bp	float	Issuer spread on the note (default 0)
collateral_spread_bp	float	Spread on the collateral leg (default 0)
issuer_call	string	"Applicable" to enable call schedule
call_dates	list	List of call dates (DD-MM-YYYY)
callable_type	string	bermudan autocallable
autocall_trigger	object	trigger_level_pct, mc_paths, mc_vol
valuation_mode	string	to_maturity to_first_call to_worst_call

Key Outputs

- selected_npv — structured note price
- npv_to_maturity, npv_to_first_call, npv_to_worst_call
- collateral_pv — standalone collateral bond value

- swap_adjustment — difference between structured and collateral leg
- cashflows — per-period detail